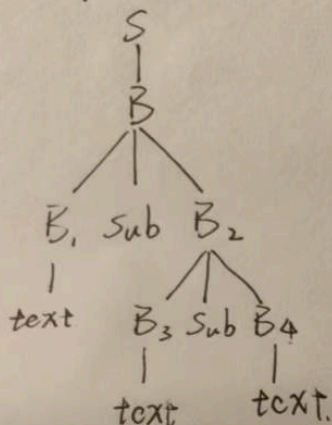


语法树如下, 为了便于区别, 标下标区分不同递归实例



为方便描述省略匹配过程。

由翻译模式可知, 最初 $B.ps = 10$ 往下进行 DFS 递归:

$B1.ps = B.ps = 10$. 继续往下, $B1.ht = \text{getHeight}(B1.ps, \text{text.lexval}) = 80$

$8 * B1.ps = 80$ ~~$B1.dp$~~ $B1.dp = \text{getDepth}(B1.ps, \text{text.lexval}) = 0$

返回上一级, $B2.ps = B1.ps \times 0.7 = 7$. 进入子树, 分别遍历得到:

$B3.ps = 7$, $B3.ht = 56$, $B3.dp = 0$

$B4.ps = 4.9$, $B4.ht = 39.2$, $B4.dp = 0$

$B2.ht = \max(B3.ht, B4.ht - B2.ps \times 0.25) = \max(56, 56)$

$B2.dp = \max(B3.dp, B4.dp + B2.ps \times 0.25) = 1.75$

然后计算 $B.ht = \max(80, 56 - 10 \times 0.25) = 80$

$B.dp = \max(0, 1.75 + 10 \times 0.25) = 4.25$

$\therefore \text{height} = 80$ $\text{depth} = 4.25$